Tanishqa Chopade

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SUMMARY

Computer Science & Engineering student at COEP Technological University pursuing a Minor in Banking, Financial Services & Insurance and a concurrent B.A. in English Literature, combining systems engineering, financial modelling, and analytical reasoning. Builds full-stack and backend systems involving REST APIs, distributed locking, immutable commit graphs, cryptographic hashing, and smart contracts. Developed a parametric crop-insurance prototype combining ground sensor telemetry and NASA MODIS satellite vegetation data with on-chain hash anchoring, alongside a version-control platform leveraging SHA-256 hashing, PostgreSQL commit histories, and Redis concurrency controls. Also engineered a clinical platform adopted across 10+ healthcare institutions. Combines technical depth with startup ecosystem leadership and international Taekwondo experience, bringing a multidisciplinary perspective to decentralised systems and verifiable real-world applications.

PROJECTS

AgriChain-Sat: Dual-Source Parametric Crop Insurance | *Smart contracts, Hardhat, Django, MQTT*

github.com/Dhruva-0206/Agri Chain

- Built a prototype where a smart contract pays a farmer only when two independent sources agree on drought: ground soil sensors and a NASA MODIS satellite vegetation (NDVI) record.
- Streamed signed readings from four simulated sensors over encrypted MQTT into a Django gateway that validates and stores them, then writes a SHA-256 fingerprint of each batch to the chain.
- Tested with Hardhat and pytest; dashboard figures are labelled live, cached or simulated; runs on a local test chain.

SheetTrack: Git-Inspired Version Control for Google Sheets | *Next.js, TypeScript, PostgreSQL, Prisma, Redis*

https://github.com/tanishqachopade/sheet-track.git

- Designed an immutable commit graph in PostgreSQL with parent-commit references, so any historical state can be rebuilt or rolled back without full copies.
- Built a cell-level diff engine using SHA-256 row hashing and HashMaps that detects added, deleted, modified, formula and reordered rows in near-linear time.
- Prevented commit race conditions with Redis distributed locking and atomic workflows; integrated Google OAuth, Sheets and Drive APIs and Gemini for AI change summaries.

Corporate Credit Risk Early-Warning & Stress Testing System | *Python, XGBoost, SHAP, FastAPI, Next.js*

- Compared Logistic Regression and XGBoost on an imbalanced Polish-company bankruptcy dataset, benchmarked against the Altman Z-score, using engineered liquidity, leverage, profitability and efficiency ratios.
- Added SHAP explanations and a dashboard that simulates revenue, profit and liquidity shocks and shows the effect on predicted bankruptcy risk.

Mini Search Engine: Ranking, Caching & Evaluation | *Python, Flask, SQLite*

- Built a QueryEngine behind both a CLI and a Flask app, with TF-IDF, BM25 and cosine scorers behind a single strategy interface and heap-based top-k retrieval (O(n log k)) for a scientific paper retrieval system.)
- Implemented a thread-safe LRU cache (HashMap + linked list) with versioned SQLite persistence, Rocchio personalisation, MMR diversification and a Precision@k/Recall@k/MRR evaluation harness.

EXPERIENCE

Engineering Intern, MedMarvel Software Solutions (MedVirtuoso) *2026*

- Replaced a manual Google Drive radiology workflow with a centralised clinical platform (Next.js, PostgreSQL, Prisma, Docker), now adopted across 10+ healthcare institutions.
- Designed multi-institution data models and hierarchical access control for 7 clinical user roles.
- Built streaming pipelines for 1.5-2 GB DICOM/PET transfers through QNAP storage and integrated a customised MRIcroGL clinical viewer.

Head of Web and App, COEP College *2025 – Present*

- Led a 10-member team building and deploying a 20+ page event platform (React, Tailwind CSS, Express.js, MongoDB) that supported 10,000+ users at launch.
- Built a QR-based digital ticketing platform with offline-capable verification for large events.

Head of Project Management, i2i, COEP Incubation Cell *2024 – 2026*

- Evaluated 500+ startup applications across seven sectors and led founder mentorship with Eaton India professionals.

Incoming Summer Intern, Hilti Technology Solutions | *Data Analytics and Software Development* *May 2027*

Incoming Winter Intern, Crisil – S&P Global | *AI Intern* *December 2026*

EDUCATION

COEP Technological University | *B. Tech, Computer Science & Engineering; Minor in BFSI* *2024 – 2028*
CGPA: 8/10

IGNOU | *B.A. (Hons) English Literature* | ***CBSE XII: 92% (2024) | ICSE X: 96% (2022)*** *2025 – 2028*

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C++, C, SQL
- **Web and backend:** React, Next.js, Node.js, Express.js, FastAPI, Django, Prisma, PostgreSQL, Redis, Docker, REST APIs, Git
- **Web3, Data and ML:** Smart contracts, Hardhat, on-chain hash anchoring, Pandas, NumPy, scikit-learn, XGBoost, SHAP.

ACHIEVEMENTS

- Represented India at the **Asian Taekwondo Championships (Jordan); multi-national taekwondo medallist and 1st Dan Black Belt**.
- Published writer: contributor to Abhiyanta, COEP's century-old magazine; self-published fiction novelist at age 14.